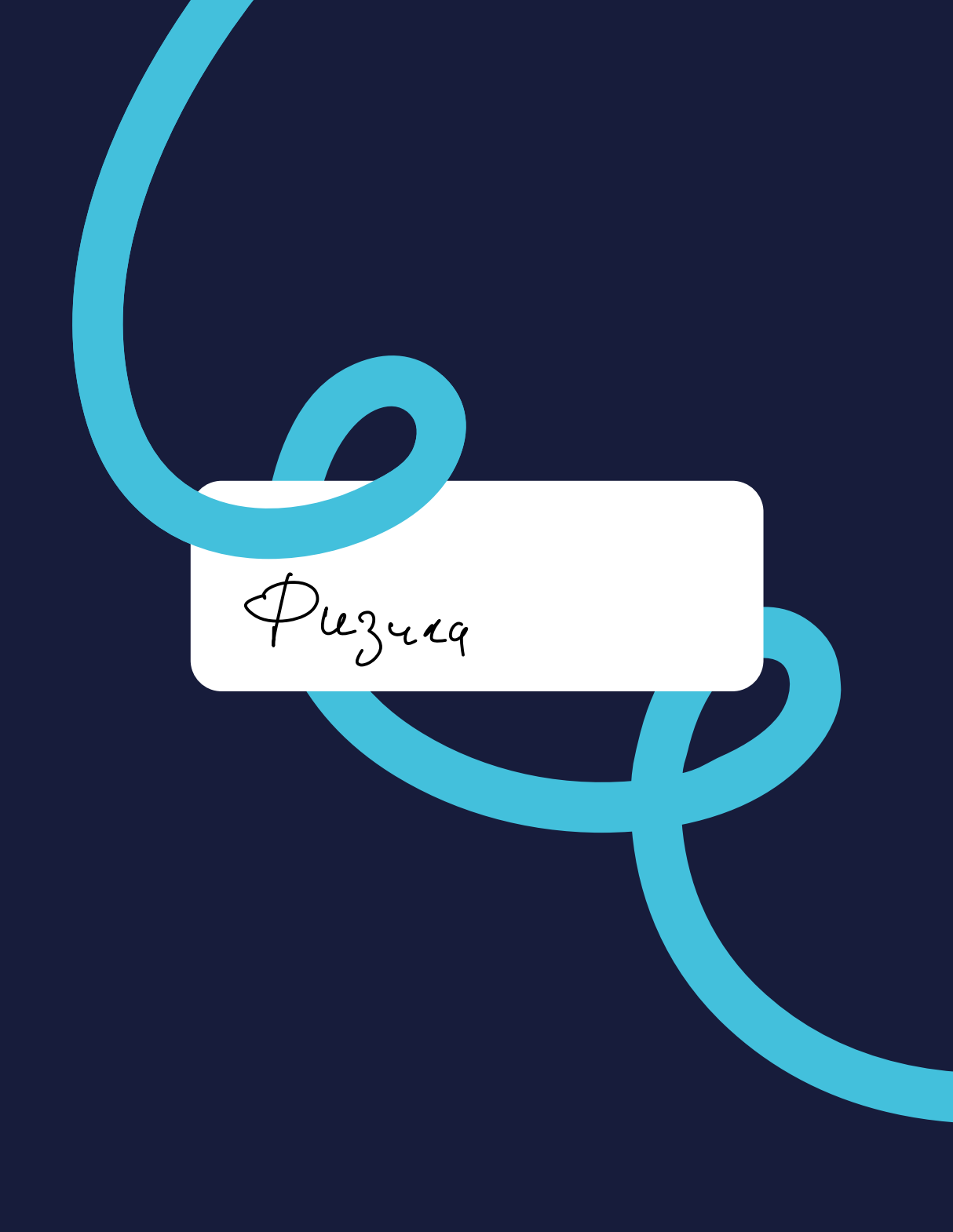


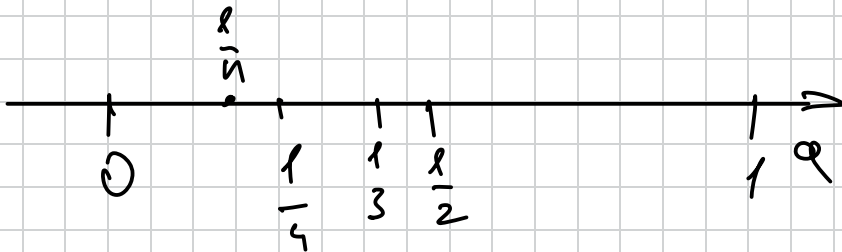


Физика

Вороков Артем Владимирович

Последовательности

$$a_n = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n} \right\}$$



a_n стремится к нулю — a_n есть предел
(сходится)

Предел — $\lim$

$$a_n \rightarrow 0$$

$$\lim_{n \rightarrow \infty} a_n = 0$$

$$b_n = 3 + \frac{2}{n}$$

$$\lim_{n \rightarrow \infty} b_n = 3$$

$$y = f(x)$$

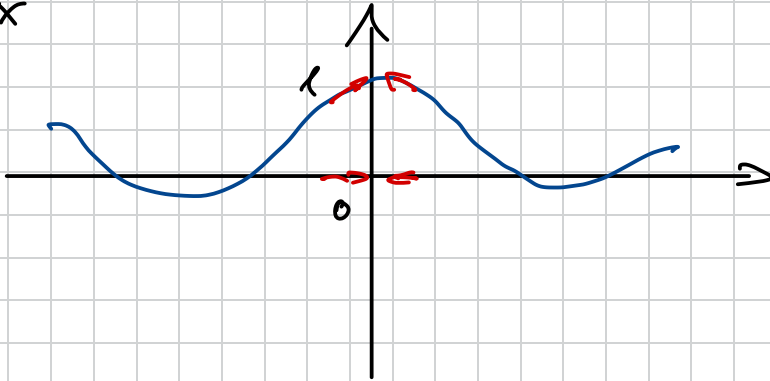
$$y = x^2$$

$$\lim_{x \rightarrow 2} x^2 = 4$$



$$y = \frac{\sin(x)}{x}$$

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$$



$$v = \frac{\Delta S}{\Delta t} = \frac{S(t + \Delta t) - S(t)}{\Delta t}$$

$$S(t) = t^2 \quad v(3) = \rightarrow 6 \text{ m/c}$$

$$1. \Delta t = 1 \text{ c}$$

$$\Delta S = S(4) - S(3) = 16 - 9 = 7 \text{ m}$$

$$v = \frac{\Delta S}{\Delta t} = \frac{7}{1} = 7 \text{ m/c}$$

$$2) 2. \Delta t = 0,1 \text{ c}$$

$$\Delta S = S(3,1) - S(3) = 0,61 \text{ м}$$

$$v = \frac{\Delta S}{\Delta t} = \frac{0,61}{0,1} = 6,1 \text{ м/с}$$

$$3. 3. \Delta t = 0,01 \text{ c}$$

$$\Delta S = S(3,01) - S(3) = 0,0601 \text{ м}$$

$$v = \frac{\Delta S}{\Delta t} = \frac{0,0601}{0,01} = 6,01 \text{ м/с}$$

$$4. \Delta t = 0,001 \text{ c}$$

$$v = 6,001 \text{ м/с}$$

$v(3) \rightarrow 6 \text{ м/с}$ (мгновенная скорость)

$$v(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta S}{\Delta t}$$

$$v(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta S}{\Delta t}$$

$$\Delta S = (t + \Delta t)^2 - t^2 = 2t\Delta t + \Delta t^2 = \Delta t(2t + \Delta t)$$

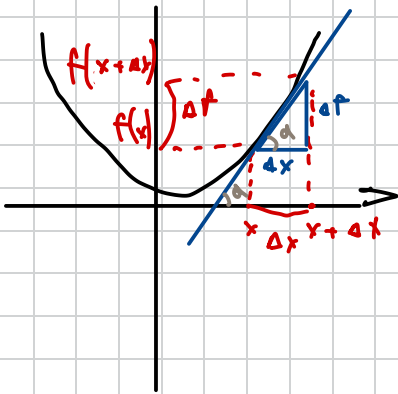
$$v(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta t(2t + \Delta t)}{\Delta t} = \lim_{\Delta t \rightarrow 0} 2t + \Delta t = 2t$$

$$v(t) = 2t$$

$$v(3) = 6$$

Производная

$$y = f(x)$$



$$\frac{\Delta f}{\Delta x} = \text{tg } \alpha$$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta x} = \text{tg } \alpha$$

Производная

